BE-04: System Context Diagram

Intelligent Kanban System with an AI Assistant

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YouTrack: DEMO-23

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1 Purpose

The context diagram (level 1 of the C4 model) shows the system as a single whole: **who** uses it, **which external systems** it interacts with, and **what data** crosses its boundary. The internal structure (frontend, backend, AI worker, databases) is described in BE-05 (Backend Component Diagram).

The diagram is based on the technology selection in BE-03 (DEMO-22).

2 Context Diagram

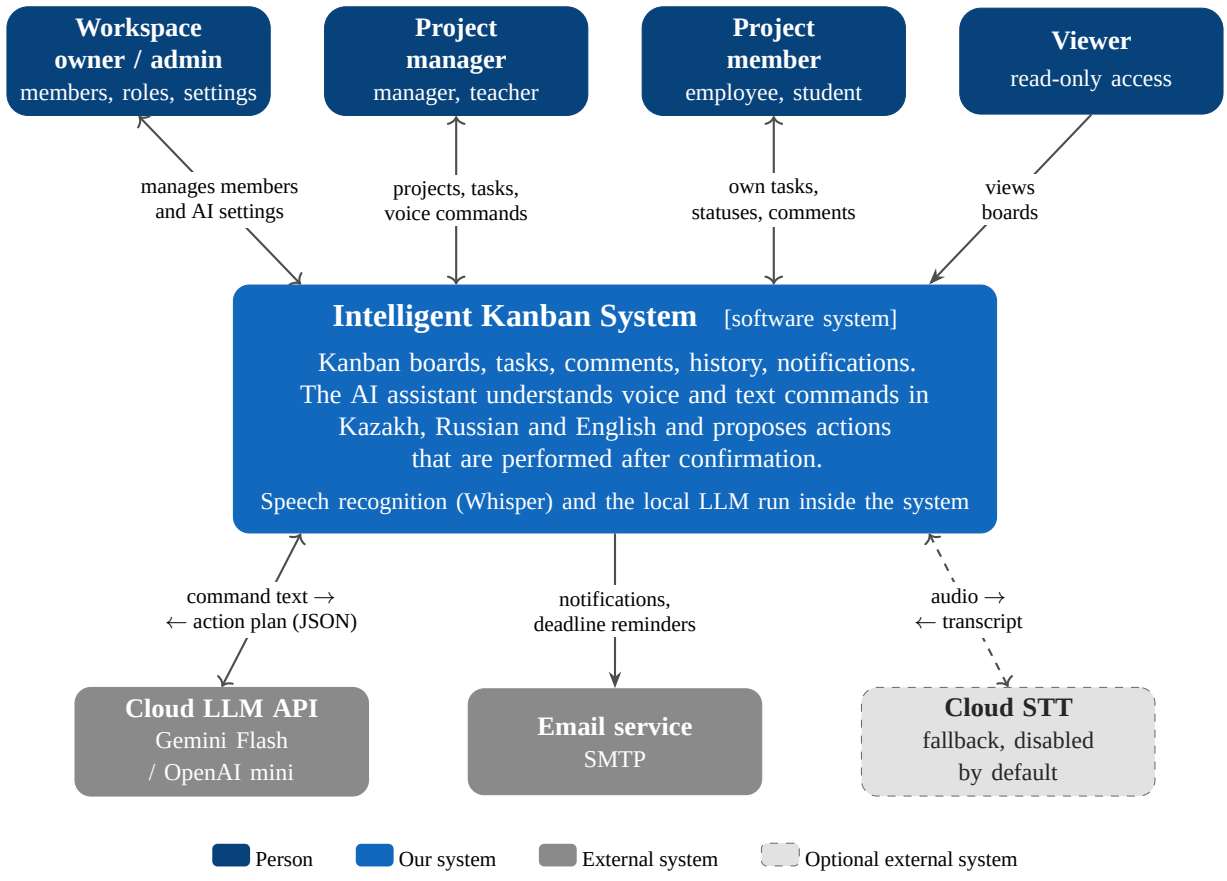

Figure 1: System context diagram (C4, level 1). Users work through a web browser over HTTPS/WSS; emails from the email service reach users' mailboxes.

3 Users

The roles follow the RBAC model from BE-03: workspace roles (owner, admin, member) and project roles (manager, member, viewer). One person can have different roles in different projects.

Role	What they do in the system	Example AI commands
Workspace owner / admin	Creates the workspace, invites members, assigns roles; enables or disables the cloud LLM (private mode) and the cloud STT fallback; views the audit log	None (managed through the UI)
Project manager	Creates projects and boards, adds members, creates and assigns tasks, sets deadlines and priorities, records meeting outcomes	«Создай задачу для Данияра: подготовить презентацию до пятницы, высокий приоритет»; “Which of our team’s tasks are overdue?”
Project member	Works on their own tasks: changes status, comments, creates tasks in accessible projects; receives notifications	«Задача по интеграции платёжной системы завершена»; «Данияға ертеңге дейін API тексеру тапсырмасын бер»
Viewer	Only views boards and tasks (for example, a teacher or a client); cannot change data	Search Tasks and Project Summary only

4 External Systems

System	Required?	Interaction	If unavailable
Cloud LLM API (Gemini Flash / OpenAI mini)	Yes in the default mode; no in private mode	Receives the command text (after speech recognition), project member names, the current date and time zone. Returns an action plan in JSON: action type, parameters, missing fields	Switch to the local LLM (Ollama); manual task management always works
Email service (SMTP)	Yes	Sends emails: task assignment, mention, upcoming and overdue deadline, workspace invitation	Emails are queued and retried; in-app notifications keep working
Cloud STT (ElevenLabs Scribe / OpenAI)	No, disabled by default	Only if an admin enables the fallback: receives audio, returns a transcript	The local STT is used; if it is also unavailable, the user types the command

Why speech recognition is not an external system. Whisper (including the Kazakh model) and the local LLM (Ollama) are deployed on our server and are part of the system. Therefore, in the default mode, voice **does not leave the system boundary**.

5 Data Crossing the System Boundary

Data	Sent to	Protection
Voice messages	Nowhere (default)	Stored in MinIO with encryption, accessed via presigned URLs, deleted after 30 days. Sent to the cloud STT only if an admin enables the fallback

Data	Sent to	Protection
Command text	Cloud LLM API	HTTPS; only the text of a single command is sent, without audio or the contents of other tasks. Not sent anywhere in private mode
Project member names	Cloud LLM API	Only the names of members of the current project, without email addresses or other personal data; needed so the model can identify the assignee mention
User email and notification text	Email service	TLS; the email contains the task title and a link, without transcripts or audio

Data that **never** leaves the system: passwords (stored as Argon2 hashes), tokens, the audit log, and the contents of tasks and comments from other projects.

6 Scope and Assumptions

- All users work through a web browser (desktop and mobile); voice is recorded in the browser. There is no separate mobile app.
- Sign-in is by email and password; external identity providers (Google, university SSO) are not part of the first version.
- Integrations with external task trackers, messengers and calendars (Jira, Telegram, Google Calendar) are not part of the first version; the architecture allows adding them later as new external systems.
- Development tools (GitHub/GitLab, YouTrack) are not part of the running system and are not shown on the diagram.
- The AI assistant never changes data on its own: every action is performed only after the user confirms it and only within the user's permissions.

7 Related Documents

- **BE-03 (DEMO-22)**: technology selection; source of the external systems and data-handling rules.
- **BE-05 (DEMO-24)**: Backend Component Diagram; breaks the "Intelligent Kanban System" box down into components.